

# Vatsal Verma

Software Developer

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## PROFESSIONAL SUMMARY

Software Developer with nearly 3 years of experience designing and developing scalable microservices using Java, Spring Boot, and Spring Cloud. Expertise in building distributed systems, implementing event-driven architectures with Kafka, and deploying applications with Docker, Kubernetes, and OpenShift. Built an AI agent with RAG and MCP integrations that reduces MTTR and ticket volume for production incidents. Proven track record in optimizing REST API performance, ensuring system security, and leading end-to-end product feature delivery.

## TECHNICAL SKILLS

**Languages:** Java, Python, SQL | **Academic:** MATLAB

**Backend & APIs:** Spring Boot, Spring Cloud, Spring Framework, Spring Security, microservices, REST, OpenAPI, OAuth2/JWT, Object-Oriented Programming, System Design, Data Structures & Algorithms

**Data & Messaging:** Kafka, Couchbase, Elasticsearch, SQL

**DevOps & Quality:** Docker, Kubernetes, OpenShift, Jenkins, Maven, Git, CI/CD, JUnit, TestNG, Jira, Confluence, Bitbucket

**AI & Methods:** RAG, hybrid retrieval (BM25 + embeddings), MCP, Google ADK, DJL, Embabel, GenAI, NLP, computer vision; TensorFlow, Keras; Agile

## PROFESSIONAL EXPERIENCE

### Software Developer Experienced

Amdocs, Pune, India

August 2023 – Present

- Developed and delivered key features within a high-scale billing ecosystem of 32+ microservices. Contributed to end-to-end feature lifecycles for major Tier-1 telecom providers, including AT&T and Bell Canada, serving tens of millions of subscribers
- Build event-driven flows and secure REST APIs (OpenAPI, OAuth2/JWT, rate limiting) with Java, Spring Boot, Spring Cloud, Kafka, and Couchbase
- Engineered an AI incident-remediation agent with RAG, hybrid BM25/embedding retrieval, and multiple MCP servers; stores and queries RAG knowledge in Elasticsearch (seed data, prior RCAs) and uses Kafka/API payloads for incident context to reduce MTTR and ticket volume
- Optimized Maven-based CI/CD on OpenShift, reducing deployment time by 20%
- Resolved 30+ CVE and OWASP issues using GenAI-assisted code review (Cursor, Copilot); owned secure production releases
- Mentor junior engineers; lead production releases and post-release support

### AI/ML Research Intern

Indian Institute of Technology (IIT-BHU), Varanasi, India

August 2021 – February 2022

- Designed facial recognition with Convolutional Neural Networks (CNNs), TensorFlow, and Keras for identification of 50,000+ individuals
- Extended models for agricultural applications by analyzing unique muzzle patterns of 100+ cattle using computer vision techniques
- Researched deep learning and image processing pipelines; managed timelines and presented findings to faculty advisors

## PROJECTS

### AI-Powered Animal Detection and Alert System

Python, CNN, Computer Vision, OpenCV, Twilio API

- Developed real-time animal detection with CNNs and OpenCV (92.68% accuracy); integrated Twilio SMS alerts with low-latency notification delivery
- Applied image preprocessing and data augmentation for robustness across lighting conditions; won WildHacks 2021 for most creative Twilio API use

### Text Summarization Application

Python, Flask, NLP, LexRank Algorithm

- Built Flask web application using NLP and LexRank for text summarization (85+% accuracy); implemented tokenization, stopword removal, and sentence segmentation

## EDUCATION

### Vellore Institute of Technology, Vellore

July 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering with specialization in IoT

CGPA: 8.22/10

## ACHIEVEMENTS AND LEADERSHIP

- Innovation Award, Amdocs:** Recognized for enhancing TestNG suite execution using Kubernetes APIs for cloud cluster integration
- WildHacks 2021:** Won prize for most creative use of Twilio API in hackathon competition
- Design Head, ISA-VIT Student Chapter:** Managed a team of 15; designed and released an application on Google Play Store